

Experiment No.	
Experiment Title	¹
Student Name	
Register Number	
Faculty	Dr. Poreddy Ajay Kumar Reddy
Submission Date	

1 Objective

State the objective of the experiment.

2 Problem Statement

Describe the problem, input, output, and objective.

3 Dataset Description

Dataset Name	
Dataset Source	
Number of Samples	
Number of Features	
Number of Classes	
Missing Values	
Train-Test Split	

4 Exploratory Data Analysis

Include all relevant visualizations.

- Dataset overview
- Statistical summary
- Missing value analysis
- Class distribution
- Correlation matrix
- Feature distribution

¹GitHub Repository: <https://github.com/username/repository>

Figure Template

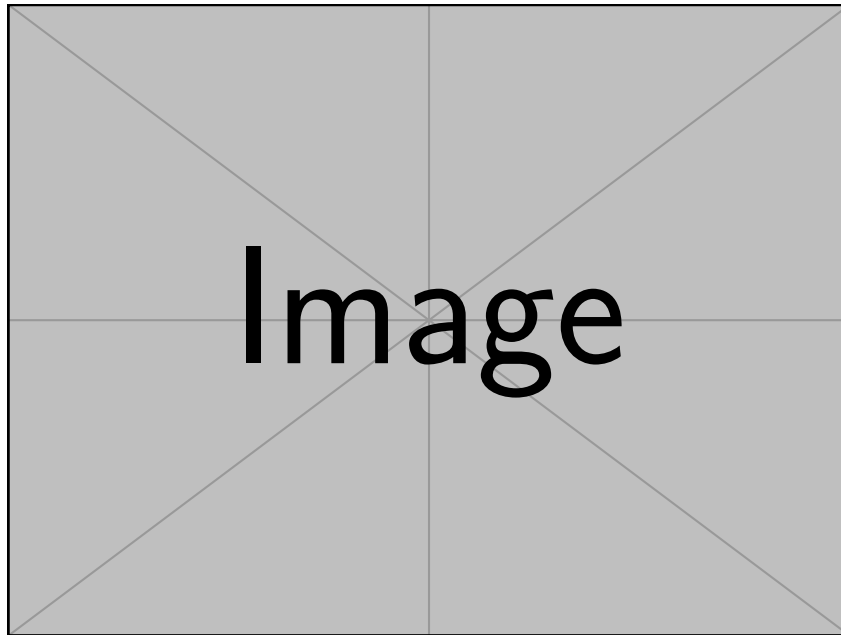


Figure 1: Replace with your figure caption

Inference:

Write 3–5 sentences explaining what the graph indicates and how it helps in understanding the dataset.

5 Data Preprocessing

Explain:

- Missing value handling
- Encoding
- Feature scaling
- Train-test split
- Feature engineering (if applicable)

6 Machine Learning Algorithm

Explain:

- Working principle
- Mathematical formulation
- Advantages
- Limitations

7 Experimental Setup

Python Version
Scikit-Learn Version
IDE
Operating System
Hardware

8 Hyperparameter Selection

Parameter	Value
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9 Results

Metric	Value
Accuracy	
Precision	
Recall	
F1-score	
ROC-AUC	

10 Result Visualization

Include all graphs generated during the experiment.

For **every graph**, provide:

- Figure
- Caption
- 3–5 line inference

Recommended figures:

1. Correlation Matrix
2. Class Distribution
3. Confusion Matrix
4. ROC Curve
5. Precision-Recall Curve
6. Learning Curve (if applicable)
7. Feature Importance (if applicable)

11 Discussion

Discuss observations, strengths, limitations and improvements.

12 Conclusion

Summarize the experiment and key findings.

13 References

Use IEEE style.